

Yashh Shenoy

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Summary — **Electronics and Embedded Systems Engineer** with hands-on experience developing and validating embedded systems for a **Formula Student EV**, spanning **VCU/ECU control**, **CAN-based data acquisition**, **BMS development and debugging**, **hardware–software integration**, and **electric powertrain testing**. Experienced across **ARM- and Atmel-based controllers**, including **STM32 and ATmega**, as well as **ESP32**, **NI myRIO/LabVIEW**, **embedded C**, multiple communication protocols, bench instrumentation, and vehicle-level fault diagnosis and validation. Expected graduation August 2027.

Skills

Languages C/C++, Python, Java

Embedded NI myRIO, STM32, ATmega, ESP32

Tools LabVIEW, MATLAB/Simulink, STM32CubeIDE, STM32CubeMX, NDrive

Protocols CAN, SPI, DMX, UART, I2C, ADC, DAC, PWM

Systems VCU, BMS, DAQ, Telemetry, Power Train

Experience

Orion Racing India – Formula Student Team, KJSCE

Feb 2024 – Aug 2026

Embedded Systems Lead & Senior Low Voltage Electronics Engineer

- Led embedded systems development for the 2026 Formula Student EV, owning the **Main Vehicle Control Unit (VCU)** firmware on **NI myRIO-1900** using **LabVIEW**; developed throttle control, Ready-to-Drive sequencing, traction control, regenerative braking, thermal pump control, real-time vehicle data acquisition, and wireless telemetry.
- Integrated wheel-speed, suspension-travel, flow-rate, NTC temperature, RPM, current, voltage, brake-pressure, rotary-position, and IMU sensors into the VCU, implementing **signal conditioning** and real-time processing for vehicle monitoring and control.
- **Runner-up, DigiKey Embedded Systems Innovation Award** at Formula Bharat 2026.
- Transitioned core VCU functionality from **NI myRIO/LabVIEW** to **embedded C** on a custom **STM32H753ZI-based VCU**, including throttle control, sensor DAQ, Ready-to-Drive sequencing, and thermal pump control; performed **SIL and HIL validation** of the control implementations using STM32 NUCLEO boards and STM32CubeIDE.
- Developed a vehicle data and communication architecture integrating **UART, CAN, and TCP**; implemented a custom CAN gateway for routing sensor data from the VCU to the vehicle CAN network and **Vector data logger**, alongside wireless vehicle-to-ground-station telemetry.
- Assisted in the development and implementation of the EV **Battery Management System (BMS)**, contributing to fault detection, CAN communication, real-time monitoring, State of Charge estimation, and passive balancing; conducted battery-pack simulations using **MATLAB/Simulink Battery Builder**.
- Led electronics debugging, testing, and validation across the vehicle, covering **low-voltage electronics, battery systems, electric powertrain, and embedded systems**; supported LV harness manufacturing and installation, electrical safety and shutdown circuitry, and resolution of hardware–software interface issues.
- Led tuning of the **BAMOCAR D3** motor controller and powertrain parameters, including PI control, I_d , I_q , ramp functions, throttle transfer functions, and field weakening, improving acceleration performance by **26.4% from 7.2 s at Formula Bharat to 5.3 s at Formula Student Germany**; performed vehicle-level simulations in IPG CarMaker to validate powertrain integration and performance.
- Developed additional vehicle telemetry using **LoRa SX1276 and nRF24**, and supported full-vehicle embedded and electrical validation during Formula Student competition deployment.
- Co-led **electrical technical inspections** for Formula Bharat, Formula Student Austria, and Formula Student Germany, and assisted with **accumulator technical inspections** for the same events.

Shenoy Studio Lighting Systems

Dec 2024 – Jan 2025

Research & Development Intern

Mumbai, India

- Assisted in the development of custom **ESP32-based control cards and MOSFET drivers** for manual and **DMX-controlled LED** lighting fixtures, implementing intensity dimming and colour-temperature control.
- Assisted in R&D of **web-based control systems for curtain motors**, working across embedded control, actuator interfacing, and system integration.
- Led development of an **ESP32-based DMX-to-TRIAC control system** for dimming halogen lighting fixtures, integrating DMX input processing with TRIAC-based AC power control.

Education

K.J. Somaiya College of Engineering, Somaiya Vidyavihar University
Bachelor of Technology in Electronics Engineering – CGPA: 9.76/10

Aug 2023 – Aug 2027
Mumbai, India

Certifications

- Automotive Skills Development Council – India – EV Safety Level 3E

Projects

Intelligent Performer-Tracking Stage Light

Final Year Project

- Developed an autonomous follow-spot system combining **UWB indoor positioning, YOLO-based computer vision, Kalman-filter sensor fusion, ESP32 control, and DMX512** to track a performer and steer a motorized moving-head light in real time.
- Designed the sensing and localization architecture using **four UWB anchors and a wearable tag** for stage-position estimation, with computer vision providing visual performer detection and identity association.
- Implemented **Kalman-filter fusion** of UWB and visual measurements to smooth noisy position estimates and predict performer motion for continuous pan/tilt control.
- Developed the embedded actuation pipeline from position estimation to **pan/tilt angle calculation, DMX mapping, ESP32 control, and RS-485 communication** with the moving-head fixture.
- Developed and tested a webcam-based prototype with closed-loop tracking, continuously adjusting fixture position from detected performer movement and tracking error.

Three-Axis Handheld Video Stabilization Gimbal

Personal

- Designed and built a low-cost three-axis handheld camera gimbal using an **Arduino Nano, MPU6050 IMU, and three SG90 servo motors**, mounted on a custom 3D-printed mechanical frame.
- Developed real-time attitude estimation and stabilization firmware using **MPU6050 DMP, I2C, PWM, and C/C++**, with yaw-pitch-roll estimation and continuous servo actuation to compensate for handheld motion.
- Implemented closed-loop stabilization with **PI-D control, integral anti-windup, and PWM slew-rate limiting** to improve transient response while limiting actuator saturation and oscillation.
- Characterized the gimbal under step disturbances, achieving **280 ms pitch and 320 ms roll settling times** within a $\pm 1^\circ$ tolerance band, with measured overshoots of 6.5% and 7.8%, respectively.

Pipelined RISC-V RV32IM Processor

Personal

- Implemented and verified **RV32I instruction types** including R-type (ADD, SUB, AND, OR), I-type (ADDI, LW, JALR), and S-type (SW, SB) instructions.
- Extended the ISA with **B-type branches** (BEQ, BNE, BLT, BGE), U-type instructions (LUI, AUIPC), and J-type instruction (JAL).
- Added the **RV32M extension** to support hardware multiplication and division instructions.
- Introduced **IF/ID, ID/EX, EX/MEM, and MEM/WB pipeline registers** and refactored the control and datapath logic for a five-stage pipeline.
- Implemented **data forwarding, pipeline stalls, and branch handling** to resolve hazards and maintain correct instruction execution.
- Performed simulation-based verification of pipeline operation and hazard handling, followed by **Xilinx Vivado simulation and FPGA hardware validation**.

Beyond Academics

- **Digital Cinematography:** Certified in Professional Approach to Digital Cinematography; avid cinephile with an interest in filmmaking and visual storytelling.
- **Music:** Electric Guitar; Trinity College London examinations up to **Grade 4**.
- **Sports:** Track Sprinting, Football, Hiking, and Trekking.